

AMIT BIKRAM ROY

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RESEARCH INTERESTS

Deep learning for computer vision: convolutional and transformer based architectures for fine grained visual classification under data scarcity, morphological classification of scientific imagery, and the interpretability of such models — particularly the detection of shortcut learning in segmented biological imaging pipelines. Secondary interests in low resource and cross lingual natural language processing for Bengali and code switched text, and in applying vision models to radio astronomical survey data. Motivated by applications to biodiversity conservation and to language technology for resource constrained settings.

EDUCATION

B.Sc. in Computer Science & Engineering

Jan 2022 to Apr 2026

Independent University, Bangladesh (IUB), Dhaka · Minor in Astronomy & Astrophysics

- CGPA: 3.54 / 4.00 (final two semesters: 3.86 / 4.00)
- ARROWS Full Funded Merit Scholarship, awarded on the basis of academic performance.
- Relevant coursework: Artificial Intelligence, Digital Image Processing, Data Structures & Algorithms, Linear Algebra, Probability & Statistics, Numerical Methods, Calculus, Database Systems, Object Oriented Programming, Mobile Application Development.

PUBLICATIONS

Peer Reviewed

[1] Roy, A. B., et al. “Machine Vision Based Extinct Fish Recognition: Bangladeshi Freshwater Case.” Proceedings of MCT4SD 2025, Springer Lecture Notes in Networks and Systems (LNNS), Scopus indexed.

Oral presentation, AIM Conference, Manila, Philippines, December 2025.

Manuscripts Completed

[2] Roy, A. B., Ghosh Nil, S., & Habib, M. T. “FishMorphHybridNet: An Efficient Hybrid Architecture for Endangered Freshwater Fish Classification.” (Manuscript completed; submission September 2026).

[3] Roy, A. B., et al. “Detecting Financial Fraud from Bengali YouTube Comments Using Cross Lingual Transformer Models.” (Manuscript completed; submission September 2026).

RESEARCH EXPERIENCE

Research Assistant, Computer Vision

Sep 2025 to June 2026

FAB Lab 360, Independent University, Bangladesh · Supervisor: Dr. Md Tarek Habib, Dept. of CSE

- Designed **FishMorphHybridNet**, a 7.47 M parameter CNN–Transformer hybrid for fine grained classification of 16 endangered Bangladeshi freshwater fish species, reaching **98.54 % test accuracy** and 0.9854 macro F1 trained entirely from random initialisation, without ImageNet pre-training.
- Established a controlled **eight architecture benchmark** under a single aligned recipe — identical splits, augmentation, optimiser, schedule, precision policy and seed — isolating architecture as the only free variable; the proposed model outperformed every lightweight and heavyweight baseline, including MaxViT-T while using **4.07× fewer parameters**.
- Identified **shortcut learning** in the strongest baseline through a Grad-CAM audit: MaxViT-T localised its evidence on segmentation mask borders and uniform background corners rather than on fish anatomy, a failure mode invisible to aggregate accuracy.
- Used Grad-CAM comparison to justify the hybrid design over its higher-accuracy CNN-only ablation, showing that the Transformer encoder produces continuous activation across the full anterior–posterior body axis where the convolutional arm relies on fragmented local texture anchors.
- Constructed and curated the underlying image corpus — acquisition, cleaning, class balancing and augmentation — for a low data regime in which no prior dataset for these species existed.
- Authored the earlier six species study (custom CNN, 8,676 field collected images, 97.42 % accuracy), published in Springer LNNS (Scopus indexed) and presented as an oral paper at the AIM Conference, Manila, December 2025.

Undergraduate Research Assistant, Astronomy & Astrophysics

Apr 2026 to July 2026

Center for Astronomy, Space Science & Astrophysics (CASSA), IUB · Supervisor: Dr. Khan Muhammad Bin Asad, Dept. of Physical Sciences

- Contributing author on the RGC II study of deep learning morphological classification of radio loud active galactic nuclei and the effect of cluster environment on the bending of radio tails. Shahal, M. S. H., ..., Roy, A. B., et al., in preparation for Astronomy & Astrophysics (ESO). Responsible for data annotation and architecture development.
- Perform expert morphological annotation of radio galaxy sources across six classes (FR0, FR1, FRII, WAT, NAT, HT), applying a fixed standard across thousands of sources to build a quality controlled training set.

- Cross match large scale radio survey catalogues (VLA FIRST, NVSS) with Astroquery and VizieR to identify bent tail radio galaxies, extract verified source metadata and resolve discrepancies between catalogues.
- Support analysis of the relationship between cluster environment and radio tail bending angle.

CONFERENCE PRESENTATIONS

- Oral presentation: "Machine Vision Based Extinct Fish Recognition: Bangladeshi Freshwater Case." AIM Conference / MCT4SD 2025, Manila, Philippines, December 2025.

PROJECTS: RESEARCH

Bengali and Code Switched Financial Fraud Detection (NLP)

2026 to Present

FAB Lab 360, IUB · Python, XLM RoBERTa, mBERT, YouTube Data API · Supervisor: Dr. Md Tarek Habib

- Building the first dedicated corpus of Bengali and "Banglish" (code switched) online comments annotated for financial scams, fraudulent investment claims and clickbait.
- Designed an active, semi supervised annotation pipeline to reduce manual labelling cost while maintaining inter annotator consistency.
- Fine tuned cross lingual transformer models (XLM RoBERTa, mBERT) for classification in a low resource, code switched setting; manuscript completed.

Brain Tumor Detection from MRI using CNNs and Transfer Learning

2025

Python, TensorFlow / Keras, VGG16 · Course project, CSE425 Artificial Intelligence, IUB

- Implemented and compared two custom CNN architectures against a fine tuned VGG16 transfer learning model for binary tumour classification on 2,065 augmented MRI scans.
- Fine tuned VGG16 reached 99.7% test accuracy against 93 to 94% for the custom CNNs; evaluated with loss curves and confusion matrices under a controlled preprocessing and augmentation pipeline.

Domain Grounded AI Tutor using Retrieval Augmented Generation

2025

Python, LangChain, FAISS, Sentence Transformers, Streamlit · Course project, Systems Analysis & Design, IUB

- Designed a RAG tutoring system that answers strictly from a student's own textbooks and past exam papers, suppressing hallucination by grounding every response in retrieved course material.
- Built the end to end pipeline covering PDF extraction, chunking, embedding, FAISS semantic search and a Streamlit chat interface, plus a module that mines past paper patterns to generate realistic practice questions.

Sustainable Heat Management for the Dhaka Urban Environment

2023

Python, geospatial analysis · Shortlisted, Green Genius Competition, Season 4

- Analysed 50 years of temperature records for Dhaka to characterise long term urban warming trends; produced geospatial heat maps and an evidence based set of mitigation recommendations presented to a competition panel.

PROJECTS: SOFTWARE

Nitya, Cross Platform Meditation Tracker

2026

Next.js, TypeScript, Cloudflare Workers, Postgres, Redis, Web Push · Progressive web app (web + mobile)

- Cross platform meditation app for web and mobile that tracks daily sessions and maintains streaks, built as an offline first progressive web app with scheduled push reminders and looping ambient audio.
- Engineered the delivery stack: service worker and IndexedDB caching for full offline use, a Redis backed scheduler for reliable reminder dispatch, and serverless Postgres for session and streak history. Live demo: nitya.amitroy257.workers.dev

RCM Claims Quality Assurance & Audit Dashboard

2026

Python, SQL, Streamlit, Plotly · Live demo: rcm-app-dashboard-app.streamlit.app

- End to end auditing tool for insurance claims data with five automated validation checks for data integrity exceptions, real time multi attribute filtering and a synthetic data fallback mode.

PGN Tree Merger

2025

Python · github.com/amitroy257/pgn-tree-merger

- Merges multi line PGN chess exports into unified opening trees, with graceful handling of malformed syntax.

Bongo Dictionary

2025

Java, SQLite · github.com/amitroy257/Bongo-Dictionary

- Offline English to Bengali dictionary with part of speech annotation and full CRUD, designed for low connectivity environments.

PROFESSIONAL EXPERIENCE

Machine Learning Engineering Intern

Jul 2026 to Present

FlyRank.AI, Chicago, Illinois, USA (Remote)

- Engineer autonomous AI agents and large scale data collection pipelines that convert unstructured web data into structured, queryable datasets.
- Implement vector databases and semantic retrieval, improving retrieval accuracy over keyword baselines by matching on meaning.
- Integrate machine learning models into production full stack web applications, from requirement through implementation to deployment.

Information Systems & Technology Operations Intern

Apr 2023 to Aug 2023

Top Jeans Ltd., Dhaka, Bangladesh

- Maintained the networked enterprise IT environment across a commercial manufacturing production floor; diagnosed and resolved hardware and software faults, reducing system downtime.

TEACHING, MENTORING & ACADEMIC SERVICE

- **Verified Chess Coach & College Ambassador, Chess.com** (Apr 2024 to Present): coach students aged 8 to 14 internationally in English; design individual training plans, track measurable progress and adapt instruction to each learner's level.
- **President, IUB Chess Club** (Mar 2023 to Apr 2024); **Student Advisor** (2024 to Present): led the executive committee, organised inter university tournaments and ran a regular training programme for members; continue to advise the serving committee.
- **Organiser, MTB International Masters Chess Tournament 2025**: coordinated operations for an International Master norm tournament while competing in it.

TECHNICAL SKILLS

Deep Learning / ML: PyTorch, TensorFlow, Keras, scikit learn; CNNs, transfer learning, transformers (XLM RoBERTa, mBERT); YOLO, OpenCV, NLTK; RAG pipelines (LangChain, FAISS, Sentence Transformers).

Scientific Computing & Data: Python (NumPy, Pandas, Matplotlib, Seaborn), SQL, Jupyter, Google Colab, Streamlit, Plotly.

Astronomy: Astroquery, VizieR, VLA FIRST / NVSS catalogue cross matching, radio galaxy morphological classification.

Software Engineering: Java, JavaScript (Node.js), TypeScript, Next.js, PHP, Django, MySQL, MongoDB, PostgreSQL, Git, Linux CLI.

Other: LaTeX, Android Studio, VS Code, Microsoft Office.

AWARDS, SCHOLARSHIPS & HONOURS

- ARROWS Full Funded Merit Scholarship, Independent University, Bangladesh (2022 to 2026), awarded for academic performance.
- Champion, Database Project Showcase, IUB Intra University Tech Fest (2023).
- Champion, NSUCEC Gamewave 2.0 (2025).
- Shortlisted, Green Genius Competition, Season 4 (2023), urban heat mitigation proposal.
- GPA 5.00 / 5.00 at both HSC (2020) and SSC (2018), Dhaka Board.

COMPETITIVE CHESS (SELECTED)

- FIDE Rated National Chess Player of Bangladesh: rating 2049; ranked among the top 14 players nationally (2024).
- Selected to represent Bangladesh at the Asian Youth Team Chess Championship.
- Second Runner up, Collegiate Chess League Spring 2026 (Chess.com), global field of 350+ universities; Division Champion, Collegiate Chess League Autumn 2023, field of 290+ universities.
- Runner up, National Inter University Chess Championship, Ministry of Youth & Sports (2022); Runner up, National Sub Junior Chess Championship (2016).

LANGUAGES & TEST SCORES

Bengali: Native proficiency.

English: Professional working proficiency. IELTS Academic 7.0 overall (Listening 8.0 · Reading 6.5 · Speaking 6.5 · Writing 6.0).

REFEREES

Dr. Md Tarek Habib, Dept. of CSE, IUB. · tarek@iub.edu.bd. Undergraduate thesis supervisor; research supervisor, FAB Lab 360.

Dr. Khan Muhammad Bin Asad, Dept. of Physical Sciences & CASSA, IUB. · kasad@iub.edu.bd. Research supervisor, RGC II project.